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An implicit storage method for 3D geological models supporting multiscale modelling and cross-sectional analysis

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ABSTRACT

To address the challenges of efficiency and flexibility in multiscale visualization and cross-sectional analysis of 3D geological modelling, this paper proposes an implicit storage and sectioning method tailored to the implicit modelling workflow. This method centres on implicit functions, storing geological interface functions and associated modelling metadata directly. It enables on-demand, rapid model reconstruction at various resolutions without repeatedly executing the explicit meshing process. Furthermore, an implicit sectioning strategy is developed on the basis of the stored implicit functions. It efficiently generates cross-sectional models through initial profile construction and sequential interface-based segmentation. This method is validated through experiments in geologically complex regions featuring faults, mined-out zones, and intrusive rocks. The results show that under multiscale modelling scenarios, the greater the modelling data volume is, the more significant the efficiency gains achieved by the proposed method. Moreover, the sectioning process notably outperforms traditional mesh-based cutting methods in terms of generation speed. In addition, the proposed method supports the generation of various types of cross-sectional geometries, such as tunnel-aligned and curved profiles. This method further extends the application potential of implicit modelling in engineering geological analysis and visualization. The relevant datasets and source code have been made publicly available to support further research and practical applications.

1. Introduction

Three-dimensional (3D) geological modelling (Houlding, 1994; Turner et al., 2021) is a process that uses computer technology to transform geological data and information into three-dimensional models, allowing intuitive and vivid visualization of underground structural features and distributions of materials. In practice, geological modelling methods are generally categorized into two main types: explicit modelling and implicit modelling. Explicit modelling typically relies on the direct definition of each object in the model, where the coordinates of key nodes, line segments, and surface patches are explicitly specified to construct stratigraphic surfaces (Ming et al., 2010; Wu, 2004). In contrast, implicit modelling generates a continuous scalar field through spatial interpolation, where geological interfaces are represented by iso-surfaces (e.g., $f(x,y,z) = 0$) (Guo et al., 2018; Martin and Boisvert, 2017). Algorithms such as marching cubes (Lorensen and Cline, 1987) and marching tetrahedra (Bagley et al., 2016) can then be applied to sample the implicit functions and extract explicit geological surfaces (Hillier et al., 2014). In this study, these extraction algorithms are regarded as surface-generation tools within the broader implicit modelling workflow, whereas the focus is on how implicit modelling results can be stored and reused before explicit mesh extraction. In contrast, implicit modelling derives implicit functions directly from raw geological data, offering strong reproducibility and suitability for automated modelling workflows. In this context, a range of methods have been proposed, including the signed distance function (SDF) (Wilde and Deutsch, 2012), radial basis function (RBF) interpolation (Cuomo et al., 2017), and kriging (Lajaunie et al., 1997). In recent years, emerging machine learning-based intelligent 3D geological modelling approaches have also been regarded as implicit modelling techniques

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(Hu et al., 2024), for example, implicit neural network (INR)-based 3D geological modelling methods (Hillier et al., 2023), which construct implicit scalar fields to represent geological structures.

Collon et al. (2015) combined implicit surfaces with a Voronoi-adaptive remeshing technique, demonstrating the practical value of implicit methods within structurally complex geological regions. Wellmann and Caumon (2018) conducted a comprehensive review of modern structural geological modelling approaches, highlighting that various techniques have been developed to incorporate observational data, geological knowledge, and uncertainty into subsurface models. However, modelling workflows are often fragmented in practice: once a model is constructed, it is typically used statically for visualization or analysis, making it difficult to refine or scale flexibly (Jessell et al., 2022; Lindsay et al., 2013). This is particularly evident in scenarios requiring multiresolution meshing and cross-sectional analysis, where the full potential of implicit modelling has yet to be realized.

Currently, most mainstream 3D geological models are defined using the boundary representation (B-Rep) method: each solid (or region) is defined by a set of surfaces, each surface is defined by a set of edges, and each edge is defined by two endpoints. As a result, commonly used model storage formats (such as OBJ and FBX) as well as geological modelling-specific formats such as RINGMesh (Pellerin et al., 2017), GMesh (Geuzaine and Remacle, 2009), and GeoSciML (Geoscience Markup Language (Sen and Duffy, 2005)), all store the complete set of vertices, surface elements, and texture information. Although formats such as ResqML (Reservoir Markup Language (Energistics, 2016)) focus on structured modelling, they fundamentally still rely on explicit meshes for model representation. These formats depend on explicit meshes, which suit rendering and finite-element analysis but struggle when the model must be shown at different levels of detail (LOD). For example, large-scale geological models often employ multiresolution encoding strategies to support interactive visualization (Sicat et al., 2023), which requires generating and storing a separate explicit mesh at each resolution level—greatly increasing the modelling time and storage burden. Even without advanced LOD techniques, viewing or simulating models at lower resolutions still requires regeneration and storage of simplified meshes—an approach that is highly inefficient in both storage and computation.

Several open-source and commercial 3-D modelling platforms adopt a similar strategy. LoopStructural (Grose et al., 2021), for example, combines RBF/finite-difference solvers and writes the potential field together with observational metadata into a single project file. GemPy (de la Varga et al., 2019) solves the scalar field on a regular grid by cokriging, whereas the commercial package GeoModeller organizes a project with XML plus externally linked explicit geometries and re-interpolates the potential field at start-up. In all these tools, the implicit solution is “hardened” into a fixed-resolution grid for storage; consequently, multiscale analysis still requires repeated interpolation or the maintenance of several meshes at different resolutions.

In theory, implicit models can reconstruct geometric information at arbitrary resolutions, enabling the decoupling of the modelling and visualization processes (Collon and Caumon, 2017). However, most existing implicit modelling tools still convert the results into explicit meshes for output. By nature, implicit modelling produces mathematical functions, which must be converted into explicit representations through sampling or mesh extraction algorithms (Caumon et al., 2013; D'Affonseca et al., 2020). Since mainstream file formats cannot directly store these implicit functions, the current workflow typically converts them into explicit meshes for storage. As a result, even when implicit modelling techniques are employed, users still need to generate and store separate meshes for each target resolution—undermining the potential of implicit modelling in multiscale modelling contexts.

Cross-sectional analysis further highlights these issues. Geological cross-sections reveal the internal structure and composition of the model (Bukovská et al., 2019; Martínez-Martínez et al., 2017; Shang et al., 2023). Traditional sectioning methods are based on explicit meshes, extracting cross-sections by computing the geometric intersections between slicing planes and mesh surfaces (Lemon and Jones, 2003; Miao et al., 2012). These algorithms rely heavily on the topological consistency and completeness of the mesh. When issues such as gaps, self-intersections, or nonmanifold edges are present, computations are prone to failure (Feito et al., 2013). Moreover, since the mesh is fixed prior to slicing, generating cross-sections at different resolutions requires remodelling to match the target level of detail. Some researchers have also proposed directly generating cross-sections from raw data, including semiautomatic or automatic derivation from borehole datasets (Hassanzadeh et al., 2022; Velasco et al., 2013) and generation from geological maps (Ran et al., 2022; Wang et al., 2023). However, such methods bypass the full 3D modelling process and do not unify the generation of models and cross-sections. As a result, the produced cross-sections are often treated as part of the input modelling data.

In summary, although extensive research has been conducted since the early 2000s on the storage and sectioning of explicit models, these approaches still fall short of meeting the practical demands of multiscale analysis on the basis of implicit modelling results. Explicit formats are incapable of representing the continuous functions underlying implicit models, requiring separate meshes to be stored for each resolution. Explicit sectioning algorithms are highly dependent on mesh quality, and multiscale sectioning requires repeated remodelling, resulting in extremely low efficiency.

To address the aforementioned issues, this paper proposes an implicit storage and cross-sectioning method specifically designed for the outcomes of implicit modelling. During the modelling stage, the method directly records the implicit function expressions and parameters of each geological interface. When visualization or cross-sectional analysis is needed, the method enables flexible generation of meshes or cross-sections at the target resolution, thus avoiding the redundancy of storing and reconstructing multiple versions of explicit meshes. The method was validated in real-world geological scenarios characterized by complex features such as faults, goaf zones, and intrusive bodies. The results indicate that the proposed method significantly improves both the efficiency and flexibility of model reconstruction and cross-section generation. It thus offers a more efficient and scalable solution for geological modelling.

2. Implicit storage

2.1. Implicit modelling

As previously mentioned, implicit modelling fundamentally involves generating mathematical functions, which are then converted into explicit mesh models through sampling or mesh extraction algorithms. A typical 3D geological implicit modelling workflow (see Fig. 1) consists of two main steps: computing the implicit functions of geological interfaces from raw data and generating explicit meshes by sampling. The specific modelling steps are as follows.

(1) Firstly, extract the coordinate and gradient data of sample points on geological interfaces from the input data. The gradient refers to the normal vector of the interface at each point.
(2) Select an appropriate implicit function and compute its parameters on the basis of the sampled interface data to derive the implicit representation of the geological interface. These functions may include interpolation-based approaches such as kriging or gradient-based implicit functions such as radial basis functions (RBF) and Hermite radial basis functions (HRBF) (Amorim et al., 2023; Guo et al., 2018).
(3) Finally, define the modelling sequence in which each geological interface function is applied (Calcagno et al., 2008), and use implicit surface extraction techniques are used to sample the implicit functions and construct the final geological model.

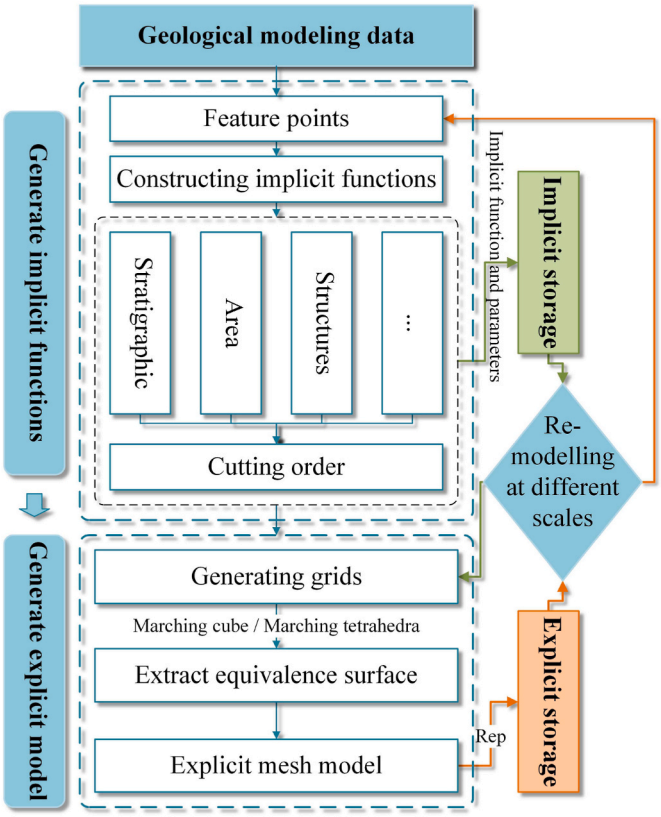


Fig. 1. Workflow of the implicit modelling.

Once the explicit mesh of a geological model is generated, traditional storage methods record all the model's vertices and surface elements. Although this approach allows for quick reloading and visualization of the model, it requires re-executing the entire modelling workflow from feature point extraction for each target resolution in multiscale modelling scenarios. An analysis of the implicit modelling process reveals that for a given dataset, the computation of implicit functions yields identical results across different modelling resolutions. Variations arise only during the meshing stage of explicit model generation. If the results of the implicit function computation, together with the necessary modelling metadata and geological processing order, can be stored, then the implicit function computation needs to be performed only once. Models at other resolutions can subsequently be generated by executing only the meshing stage.

2.2. Implicit storage of models

Existing geological model storage and exchange approaches include general geoscience standards such as GeoSciML, reservoir modelling standards such as RESQML, modelling frameworks such as Loop-Structural and GemPy, and mesh-oriented formats or libraries such as Gmsh and RINGMesh. These approaches support geological feature description, model exchange, project reconstruction, and explicit mesh representation. However, they are mainly designed for geological objects, model properties, structured grids, scalar fields, or explicit geometries, rather than for preserving reusable implicit interface functions together with their computational context and geological processing order. Therefore, the metadata defined in this study is not intended as an extension or replacement of existing standards or frameworks. Instead, it is designed specifically for the proposed implicit storage workflow and should be understood as a method-specific description for reconstructing implicit modelling results.

Following this motivation, on the basis of the entire implicit modelling workflow, implementing implicit storage for 3D geological models requires more than simply storing the results of the implicit functions. It is also necessary to store the implicit function results for each geological interface, along with additional modelling metadata. The storage of implicit functions should include both the computed parameters and the original implicit point data; the modelling metadata should cover basic information such as model boundaries and depth, as well as representation-related data including regional interfaces and the interface cutting order.

2.2.1. Storage modelling information

Traditional model storage methods are based on explicit meshes, where each point has a precisely defined coordinate. The implicit storage method proposed in this paper directly stores the implicit functions, which, unlike explicit mesh storage, do not have inherently defined spatial bounds. Therefore, to ensure that the stored results can be used to reconstruct a model over the same spatial extent, it is necessary to first store modelling metadata, including model boundaries, depth, and surface (topographic) data.

In the modelling process, surface data are typically derived from the regional digital elevation model (DEM), from which a Delaunay triangulation is constructed to generate a triangulated irregular network (TIN) that conforms to the terrain surface. In this study, we store both the TIN of the surface and its corresponding implicit function. The implicit function uses each DEM point as an implicit feature point and is represented using the same type of implicit function as that used for stratigraphic surfaces.

The next step involves storing the implicit functions generated during the modelling process. Throughout the implicit modelling workflow, various types of implicit functions are produced, including stratigraphic interface functions, regional boundary functions, and structural interface functions. These different types of implicit functions are treated differently in the subsequent explicit model generation stage, often requiring different cutting orders and processing methods. Therefore, identifiers are required to distinguish between different categories of implicit functions.

A well-defined implicit function consists of both a function expression and its associated parameters. Therefore, storing an implicit function requires the preservation of both its expression and parameter values. However, the expressions of implicit functions are often complex, making their storage and retrieval challenging to implement. As a result, identifiers are also introduced to represent the type of implicit function used. For example, the Hermite radial basis functions (HRBF) formulation (Eq. (1)) comprises a basis function, a set of known points, and a set of parameters. Thus, for this type of implicit function, a unique identifier is required to indicate that it is an HRBF based on a specific radial basis function, along with the storage of original input points and the computed parameters α and β.

$$f(x) = \sum_{i=1}^{n} (\alpha_i \varphi(x - x_i) - \langle \beta_i, \nabla \varphi(x - x_i) \rangle) \quad (1)$$

2.2.2. Storing explicit information

Once geological implicit surfaces have been generated, interfaces commonly intersect, as illustrated by surfaces E and H in Fig. 2(a). If, according to geological chronology, surface E was formed after surface H, then surface E should truncate surface H during explicit mesh construction, which may be implemented using hard or soft truncation methods (Wellmann and Caumon, 2018). Direct computation of the intersection points between implicit surfaces corresponds to the hard truncation approach, but this method often introduces numerical errors due to floating-point approximations, which may result in geometric inconsistencies. To address this issue, this study adopts a geological body peeling method based on stratigraphic deposition order (Guo et al., 2021), in which the tetrahedral mesh is sequentially cut from the youngest to the oldest geological units according to their deposition

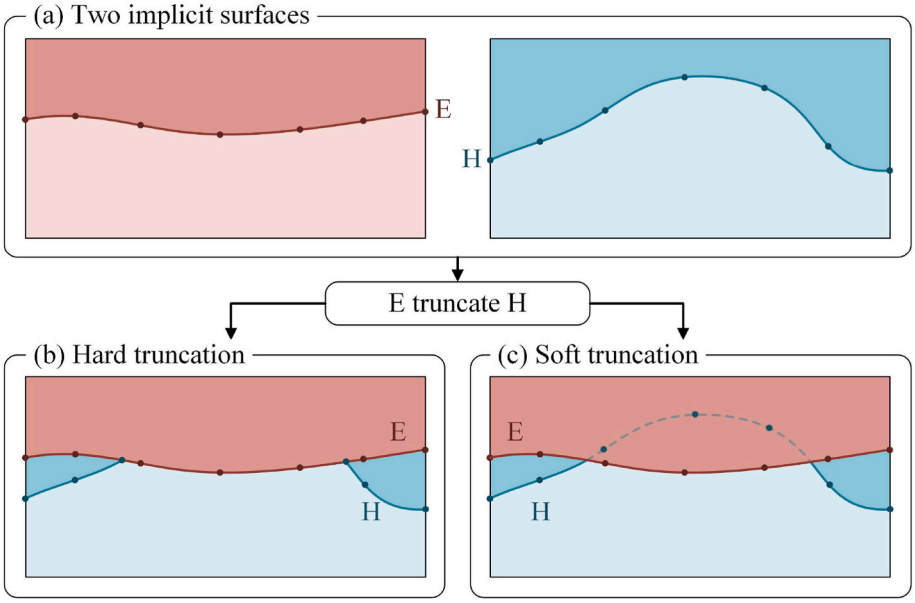


Fig. 2. Two implicit face truncation methods (adapted from Wellmann and Caumon, 2018).

order (Thiele et al., 2016).

However, regardless of the method used to construct the explicit mesh, the sequence in which geological interfaces are processed is critical to the modelling process. During the explicit model generation process, structural and stratigraphic implicit surfaces typically require different cutting orders and processing strategies. Therefore, it is necessary to store these sequences separately and use identifiers to distinguish the cutting order of each function surface. In addition, Boolean operations between implicit functions are often employed to represent complex structural configurations during structural modelling. Consequently, when storing structural implicit surfaces that involve Boolean operations, special identifiers must be added to indicate whether the function surfaces are combined and, if so, which surfaces are involved. This ensures that complex structural geometries can be accurately reconstructed during the modelling process.

3. Cross-section generation on implicit surfaces

On basis of the implicit model storage method described above, this study implements an implicit cross-sectioning method that directly generates cross-sectional models from the stored implicit functions. The corresponding workflow is illustrated in Fig. 3.

3.1. Generating the initial section mesh

3.1.1. Vertical section

Vertical cross-sections are the most common cross-sectional shapes in the cutting analysis of geological models and can be categorized into straight-line cross-sections, polyline cross-sections, and curved cross-sections on the basis of their horizontal shape.

The straight-line section is taken as an example. First, the two endpoints $A(x_1, y_1)$ and $B(x_2, y_2)$ are selected. Then, on the basis of the endpoint coordinates, the parametric equation (Eq. (2)) of the straight line can be calculated:

$$\boldsymbol{r}(t) = \boldsymbol{A} + t(\boldsymbol{B} - \boldsymbol{A}), t \in [0, 1] \tag{2}$$

where $\boldsymbol{A}$ and $\boldsymbol{B}$ are the vector representations of the endpoints and where $\boldsymbol{r}(t)$ represents the coordinates of any point on the straight-line profile, with t as the parameter.

Next, within the range of parameter t, the line segment is refined using a specified discretization step size ΔL to generate the coordinates

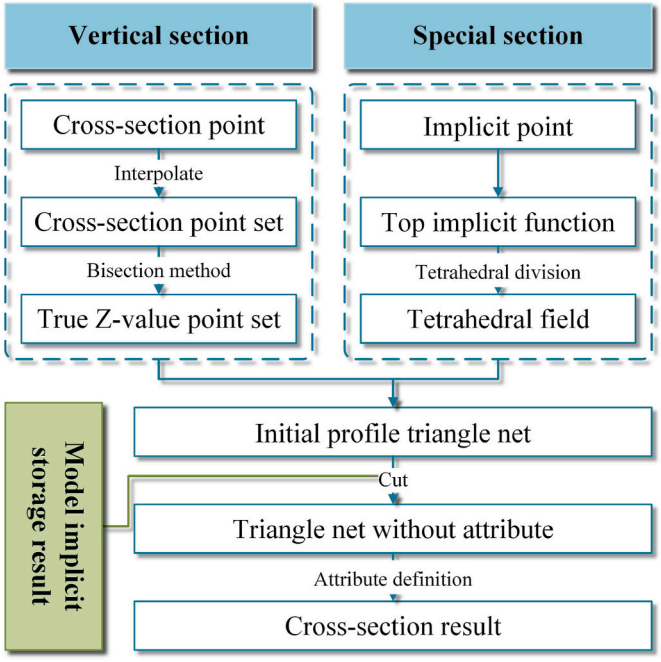


Fig. 3. Workflow of the implicit cutting process.

of intermediate points. First, calculate the length L of the line segment AB, then compute the number of intermediate points n to be inserted using the formula $n = L/\Delta L$, and finally, calculate the parameter increment $\Delta t = 1/n$. Substituting Δt into Eq. (3) yields a set of points along line segment AB with spacing ΔL.

$$\boldsymbol{r}(t_i) = \boldsymbol{A} + t_i(\boldsymbol{B} - \boldsymbol{A}), t_i = i \cdot \Delta t \tag{3}$$

where Δt represents the step size; i is an integer whose range is determined by Δt; and t_i is the parameter of the i-th point.

The method for calculating Δt is as follows: first, the length L of line segment AB is calculated; then, the number of points n from $L/\Delta L$ is calculated on the basis of the set along-line accuracy ΔL; and finally, Δt is calculated from the formula $\Delta t = 1/n$.

Finally, the initial profile mesh is generated, as shown in Fig. 4.

(1) Using Equation (6), all refined points along the section line are generated to form the set of section points $\boldsymbol{P} = \{P_{l1}, P_{l2}, \ldots, P_{li}\}$, as shown in Fig. 4(a).
(2) The actual elevation of each point is computed using the bisection method by the top surface function stored in the implicit model. This yields an adjusted point set conforming to the terrain, as shown in Fig. 4(b).
(3) Taking two adjacent points (e.g., P_{l1}, P_{l2}) as an example, project them to the target cross-sectional depth D to obtain points P_{t1}, P_{t2}. Connect segments $P_{l1}P_{t1}$ and $P_{l2}P_{t2}$. On the basis of the specified number of mesh layers n, calculate the vertical interval ΔD, and divide each segment uniformly. The resulting points are then connected to form a triangulated mesh, as illustrated in Fig. 4(c).
(4) Repeat step (3) for the remaining points in the set to complete the initial cross-section mesh, as shown in Fig. 4(d).

A polyline cross-section can be viewed as multiple straight-line cross-sections that are connected. First, a set of continuous bending points $(P_{b0}(x_{b0}, y_{b0}), P_{b1}(x_{b1}, y_{b1}), \ldots, P_{bn}(x_{bn}, y_{bn}))$ within the modelling range is selected on the X-O-Y plane, with each pair of adjacent points defining a straight line segment. Then, on the basis of coordinates of the endpoints of each segment, the parametric equation (Eq. (4)) of the line is calculated:

$$\boldsymbol{r}_i(t) = \boldsymbol{P_{bi}} + t(\boldsymbol{P_{bi+1}} - \boldsymbol{P_{bi}}), t \in [0,1], i = 0, 1, 2, \ldots, n-1 \tag{4}$$

where $\boldsymbol{P_{bi}}$ and $\boldsymbol{P_{bi+1}}$ are the vector representations of the endpoints and where $\boldsymbol{r}_i(t)$ represents the coordinates of any point on the i-th segment, with t as the parameter.

Next, each line segment is densified at a certain step size to generate the coordinates of each point (Eq. (5)).

$$\boldsymbol{r}_i(t_j) = \boldsymbol{P_{bi}} + t_j(\boldsymbol{P_{bi+1}} - \boldsymbol{P_{bi}}), t_j = j \cdot \Delta t \tag{5}$$

Finally, the initial mesh of the polyline cross-section is generated using the same method, as shown in Fig. 5.

The key to generating the initial mesh for the curved cross-section is constructing the parametric function of the curve on the X−O−Y plane. In this work, the cubic spline method is used to construct the planar curve function. The specific method is as follows. First, a set of control points $P_{c0}(x_{c0}, y_{c0}), P_{c1}(x_{c1}, y_{c1}), \ldots, P_{cn}(x_{cn}, y_{cn})$ is selected within the modelling domain. Then, for each segment, the parametric equation is computed on the basis of coordinates of its endpoints and the given constraints (Eq. (6)):

$$\boldsymbol{r}_i(t) = a_i + b_i t + c_i t^2 + d_i t^3, t \in [0,1], i = 0, 1, 2, \ldots, n-1 \tag{6}$$

where a_i, b_i, c_i, d_i are coefficients to be determined and where $\boldsymbol{r}_i(t)$ represents the coordinates of any point on the i-th segment, with t as the parameter.

The known conditions include interpolation conditions (Eq. (7)), continuity conditions (Eq. (8)), and boundary conditions (Eq. (9)).

$$\boldsymbol{r}_i(0) = P_{ci}, \boldsymbol{r}_i(1) = P_{ci+1} \tag{7}$$

$$\boldsymbol{r}_i'(1) = \boldsymbol{r}_{i+1}'(0), \boldsymbol{r}_i''(1) = \boldsymbol{r}_{i+1}''(0) \tag{8}$$

$$\boldsymbol{r}_0''(0) = 0, \boldsymbol{r}_n''(1) = 0 \tag{9}$$

Next, each curve segment is densified at a certain step size Δt to generate the coordinates of each point (Eq. (10)).

$$\boldsymbol{r}_i(t_j) = a_i + b_i t_j + c_i t_j^2 + d_i t_j^3, t_j = j \cdot \Delta t, j = 0, 1, 2, \ldots, \frac{1}{\Delta t} \tag{10}$$

Finally, the initial mesh of the curved cross-section is generated using the same method, as shown in Fig. 6.

3.1.2. Special profile

In the analysis of model cuts, in addition to common vertical profiles, it is sometimes necessary to generate profiles of special shapes. Special profiles are defined by the spatial shapes of function surfaces.

If only a set of spatial control points is available and no explicit function is provided, an implicit surface is first derived from the control points using an implicit function method, followed by the generation of the initial cross-section mesh using the marching cubes algorithm (Bagley et al., 2016).

The detailed process is illustrated in Fig. 7, which shows the generation workflow of the spherical cross-section.

3.2. Cross-sectional cutting

After the initial triangulated mesh of the cut surface is obtained, it is necessary to partition and attribute the profile mesh correctly according to the stratigraphic interfaces and properties to generate the final profile. This study adopts the approach of implicit surface-based tetrahedral segmentation proposed by Guo et al. (2021) and implements an intersection algorithm between implicit surfaces and triangles. The method of cutting a triangle with an implicit surface is equivalent to the method of cutting a triangle with a plane, as described by Chalasani et al. (1991).

In brief, the relationship between a triangle and an implicit surface is determined by evaluating the function values at the triangle's three vertices. For triangles that intersect the surface, the intersection points are recalculated, and the triangle is retriangulated accordingly to achieve the surface-based cut, as shown in Fig. 8. The new triangles must not only maintain consistent orientation but also comply with the local optimisation procedure (LOP) (Lawson, 1972) to ensure optimal triangulation, as illustrated in Fig. 8(b).

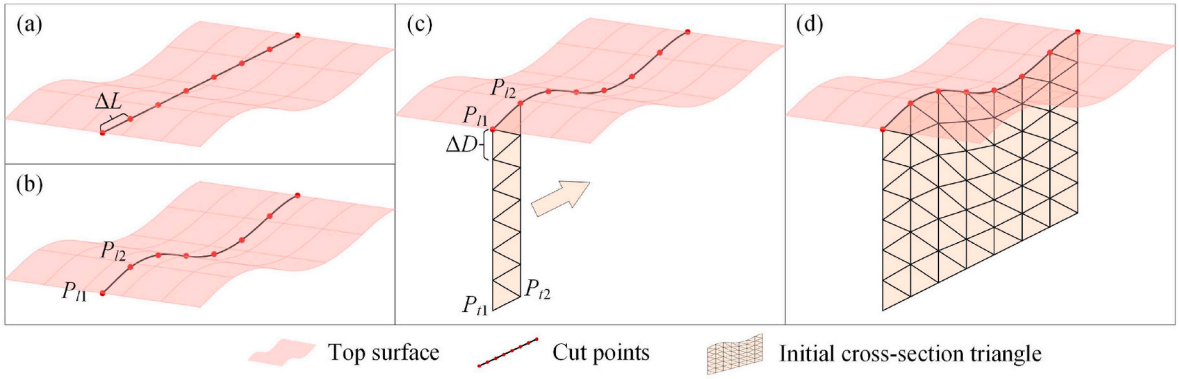


Fig. 4. Workflow for generating the initial mesh for the linear cross-section. (a) Section line refinement based on step size; (b) section line adjusted to match the terrain surface; (c) initial mesh between two adjacent points; (d) complete initial cross-sectional mesh.

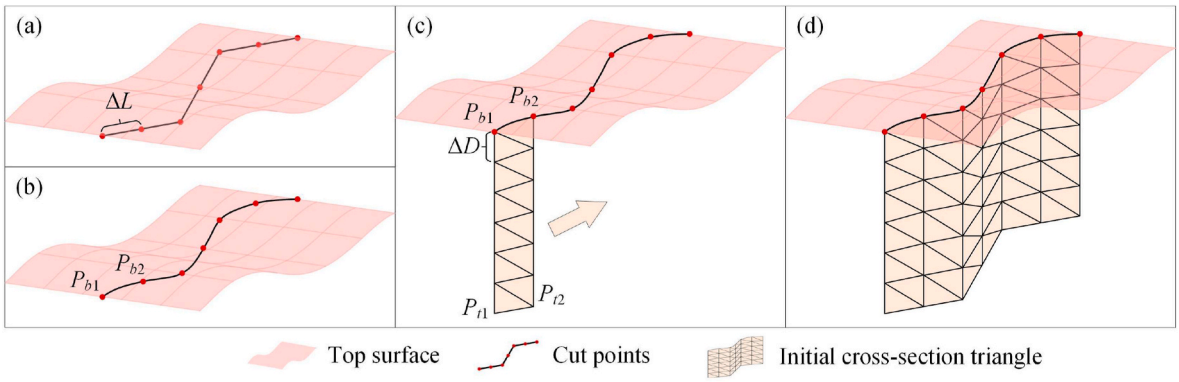


Fig. 5. Workflow for generating the initial mesh for the folded cross-section. (a) Step-based refinement of each polyline section; (b) polyline section adjusted to match the terrain surface; (c) initial mesh between two adjacent points; (d) complete initial cross-section mesh.

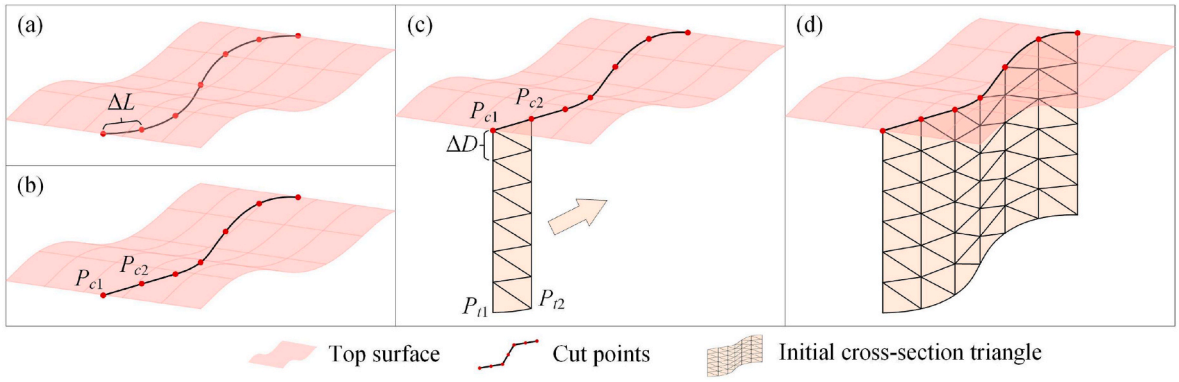


Fig. 6. Workflow for generating the initial mesh for the curved profile. (a) Curved section line refined on the basis of step size; (b) curved section line adjusted to match the terrain surface; (c) initial mesh between two adjacent points; (d) complete initial cross-section mesh.

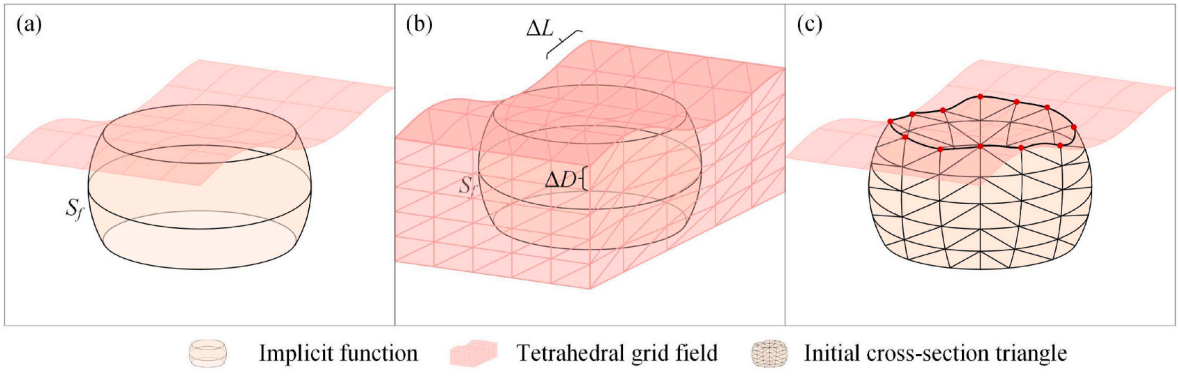


Fig. 7. Workflow for generating the initial mesh for the spherical row profile. (a) Functional representation of the spherical cross-section; (b) tetrahedral mesh field within the modelling region; (c) initial mesh of the spherical cross-section.

After performing the intersection between the implicit surfaces and the cross-section mesh, all types of implicit surfaces stored in the implicit model are sequentially applied to the cross-section mesh according to their recorded peeling order, with stratigraphic attributes assigned accordingly. This results in the final generation of the cross-section (Fig. 9).

4. Case study

On the Basis of the above implicit model storage and cross-sectioning methods, this study uses the C++ language in combination with the framework of the Qt 5.9.1 (2024, https://www.qt.io/, last access: 13 July 2024) application to implement the implicit storage of regional complex geological models and the profile generation process on the basis of implicit storage. The hardware and software environment parameters are AMD Ryzen9 5900HX 3.3 GHz CPU, NVIDIA GeForce RTX 3060 GPU, 16 GB RAM, and Windows 11 (64-bit).

The proposed method was applied to a geologically complex real-world region, which includes geological features such as intrusive rocks, goaf zones, and faults. The modelling dataset consists of 43 borehole data (including intrusive rock records), 2 goaf zone data, and 6 fault data, as shown in Fig. 10.

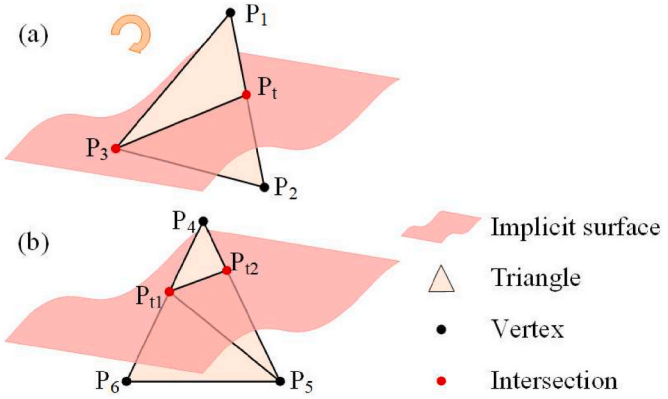


Fig. 8. Triangular cutting: (a) with one point of the triangle on the surface; (b) with no points of the triangle on the surface.

4.1. *Implicit storage of the model*

In this study, a Hermite radial basis function (HRBF)-based implicit 3D geological modelling method was used to construct the geological model of the study area, and accordingly, the implicit storage of the model was also implemented using HRBF-based implicit functions. The implemented storage format is detailed in Appendix 1. Different types of implicit functions are stored separately, including the top surface function (TOPSSURF), regional interface functions (REGIONSURF), karst cavity interface functions (CAVESURF), goaf interface functions (GOAFSURF), fault surfaces (FAULTSURF), intrusive rock interfaces (INSTRUSURF), and stratigraphic surfaces (STRATUSURF).

After the model was constructed, the generated implicit surfaces were stored according to the proposed format. The results are shown in Table 1, which presents only the stored cutting order of the implicit surfaces.

A dedicated software tool was also developed in this study, which reads the stored HRBF-based implicit model data and reconstructs the corresponding 3D model at a specified meshing resolution. A comparison between the originally constructed model (Fig. 11(a)) and the model reconstructed from the implicit storage (Fig. 11(b)) shows no difference in geometry or even in the number of triangular mesh elements, demonstrating that the proposed implicit storage method can preserve the full integrity of the 3D model. Moreover, since this method stores the implicit functions of the model, alternative surface sampling methods can be applied after loading the stored data to generate the explicit mesh. For example, the point-by-point scanning method (Meyer et al., 2008) used in Fig. 11(c).

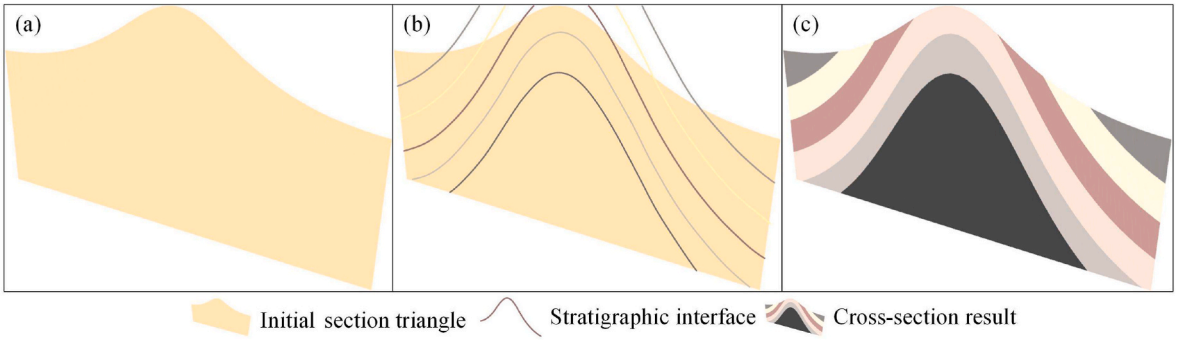


Fig. 9. Definitions of geological attributes. (a) Initial cross-sectional triangle; (b) stratigraphic interface cutting with assigned attributes; (c) cross-sectional result.

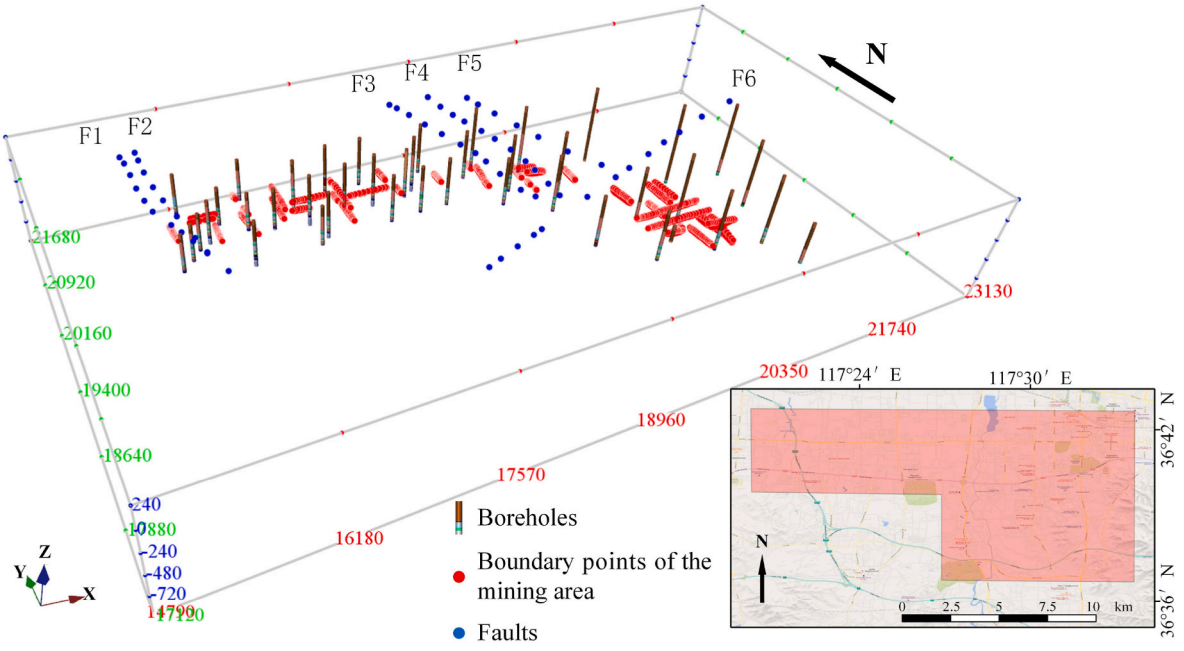


Fig. 10. Modelling data of the study area.

Table 1
Implicit storage of results (only the cutting order in the storage results is shown).

Cutting order	Order of the results
GOAF_ORDER	1, 2; 3, 4; 5, 6; 7, 8; 9, 10; 11, 12; 13, 14; 15, 16
CAVE_ORDER	
FAULT_ORDER	1, 2, 3, 4, 5, 6
REGION_ORDER	1, 2, 3, 4, 5, 6
REGION_NUM	7
REGION_ORDER_INSTRU	1, 2, 3, 4, 5, 6
REGION_ORDER_STRATU	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26

4.2. Analysis of the rebuild efficiency

As noted in the introduction, the proposed method offers a more efficient solution for multiscale modelling tasks; therefore, the efficiency analysis is carried out under the premise of multiscale modelling requirements. With conventional explicit mesh-based storage methods, changing the modelling resolution requires re-executing the entire implicit modelling process. In contrast, the proposed method stores the components common to all resolutions (implicit functions) to improve the efficiency of multiscale modelling. To verify the practicality of the method, the modelling times are compared—with and without the implicit storage strategy—under various modelling resolutions.

First, with the number of boreholes fixed at 43, a comparison is made between different modelling resolutions in terms of modelling time with

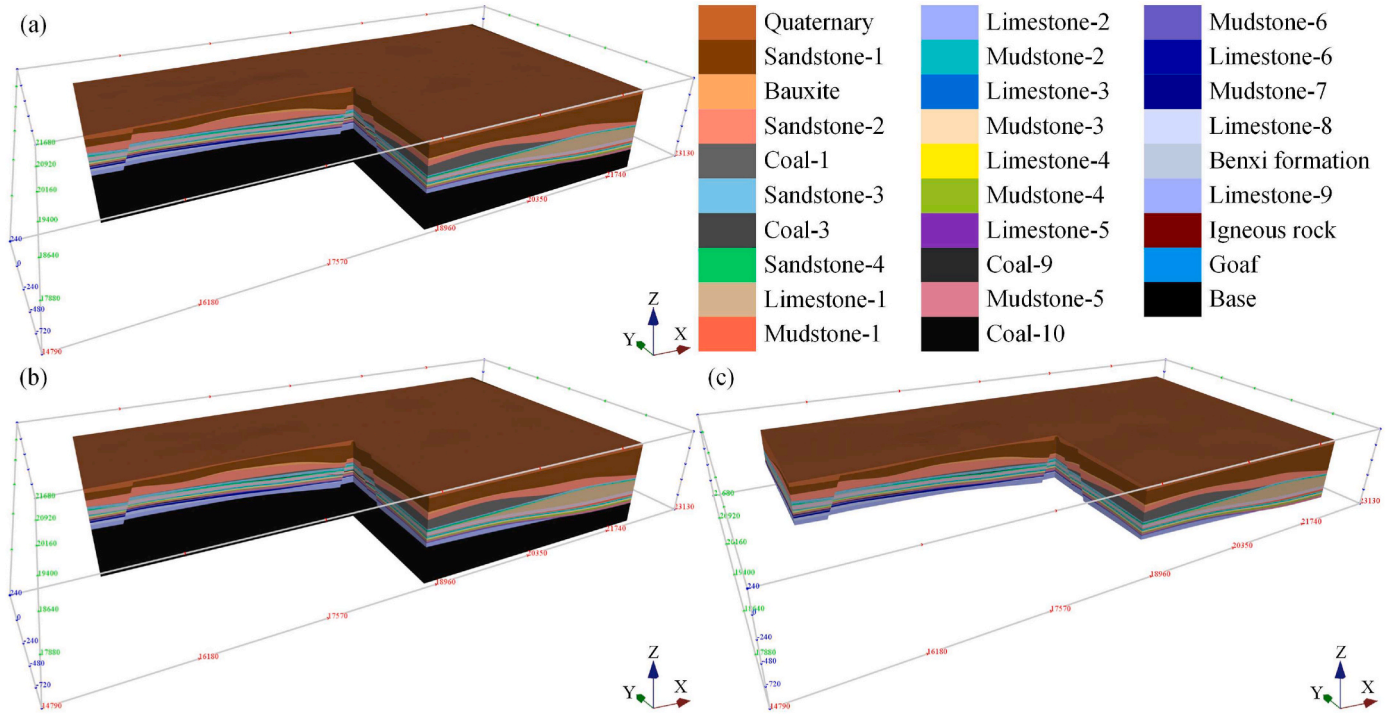


Fig. 11. Results of the model reconstruction based on implicit storage: (a) implicit modelling results (number of triangles: 5553488), (b) Model rebuilt from implicit storage results (number of triangles: 5553488), (c) Model rebuilt from implicit storage results using the point-by-point scanning method.

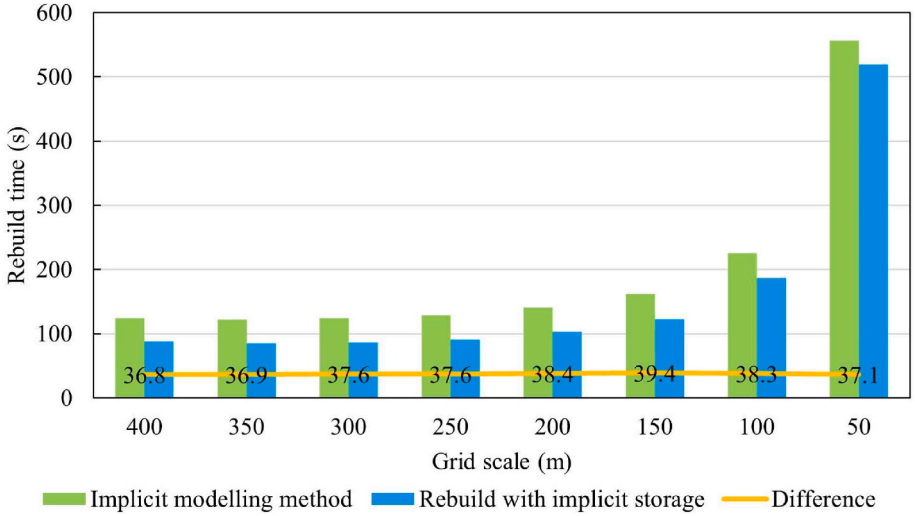


Fig. 12. Comparison of modelling time and reconstruction time from implicit storage at different resolutions with a fixed number of boreholes.

and without the use of implicit storage. Experiments were conducted under eight different mesh resolutions, namely, 400 m, 350 m, 300 m, 250 m, 200 m, 150 m, 100 m, and 50 m, and the results are presented in Fig. 12.

According to the results shown in Fig. 12, the modelling time when the implicit storage method is used is consistently lower across all resolutions. However, the amount of time saved remains nearly constant, corresponding to the time required to compute the implicit functions. This further confirms the point made in Section 2.1—that the implicit function computation step remains unchanged when modelling the same dataset at different resolutions.

To further evaluate the potential of implicit storage, a comparison was conducted of modelling time under varying data volumes at a fixed resolution of 100 m. In this experiment, the amount of modelling data was proportionally increased by duplicating and offsetting the original borehole dataset. As a result, faults and goaf zones—absent from the borehole data—were not included in the modelling process. Experiments were conducted using borehole counts of 43, 86, 129, 172, 215, 258, 301, and 344, and the results are shown in Fig. 13.

The previous experiment demonstrated that the time saved by using the implicit storage method corresponds to the time required to compute the implicit surfaces. Therefore, the percentage improvement in efficiency observed when implicit storage is used (represented by the line graph in Fig. 13) also reflects the proportion of time spent on implicit surface computation in the overall modelling process. As shown in Fig. 13, with increasing amounts of modelling data, the proportion of time spent on computing implicit functions increases within the total modelling workflow. This findings indicate that the proposed implicit storage method has strong potential for applications involving large-scale, multiresolution geological modelling.

4.3. *Implicit cutting based on implicit storage*

On the basis of the implicit storage results of the model, a fast cross-section generation workflow (i.e. implicit cross-sectioning) was also implemented. A sectioning experiment was conducted for the study area, as shown by borehole-aligned cross-section A–B in Fig. 14.

According to the results of the borehole cross-section, except for the goaf zones not captured in the borehole data, the cross-section generated by the proposed implicit cross-sectioning method matches the stratigraphic information revealed by the boreholes. Similarly, the implicit cross-sectioning method has clear advantages for multiscale cross-sectional analysis of the model, as only two parameters—the step size ΔL and the number of mesh layers n—need to be adjusted during initial section generation to rapidly generate a denser and more continuous cross-section model, as illustrated in Fig. 15.

A comparative experiment was then conducted against the explicit cross-sectioning method based on dynamic quadtree indexing (Dai et al., 2020). Considering the applicability of the explicit method, complex geological structures were excluded from this experiment. The resulting cross-sections are shown in Fig. 16, which reveals that both methods produce consistent stratigraphic layering, although the mesh generated by the implicit method is more regular and structurally coherent.

An efficiency comparison experiment was then conducted, under mesh resolutions of 100 m and 50 m. For the implicit cross-sections, the step size ΔL was also set to 100 m and 50 m, with $n = 30$ layers. The efficiency results are presented in Table 2, which shows that the proposed implicit cross-sectioning method outperforms the explicit method in both resolution settings. On average, it achieved an improvement of over 98%. Furthermore, when multiscale section analysis is needed, the explicit workflow must first create separate mesh models for each resolution, whereas our method can generate cross-sections at multiple resolutions directly, yielding an even greater efficiency gain.

Additionally, the proposed implicit cross-sectioning method supports the generation of cross-sections with complex or nonstandard geometries. Using this method, tunnel-aligned cross-sections along the metro line in the study area were generated, as shown in Fig. 17. These tunnel-based cross-sections enable more targeted geological analysis around the metro alignment. When combined with vertical cross-sections along the metro line, a clearer and more intuitive understanding of the geological conditions can be achieved. This, in turn, facilitates more informed decision-making for metro engineering and construction.

5. Discussion

In this study, we propose an implicit storage and sectioning method for 3D geological models on the basis of the principles of implicit modelling. The proposed method directly stores intermediate results—namely, implicit functions representing geological interfaces—and uses them to enable multiscale model reconstruction and flexible cross-sectional analysis. Two sets of experiments were conducted to validate the approach, demonstrating that the proposed method significantly improves both the efficiency and adaptability compared with traditional explicit storage and sectioning workflows. This is particularly evident when handling modelling tasks across varying mesh resolutions. In addition, we developed software capable of reading the implicit storage results and automatically generating models at different mesh resolutions. Both the software source code and the implicit storage data used in the case study have been made publicly available.

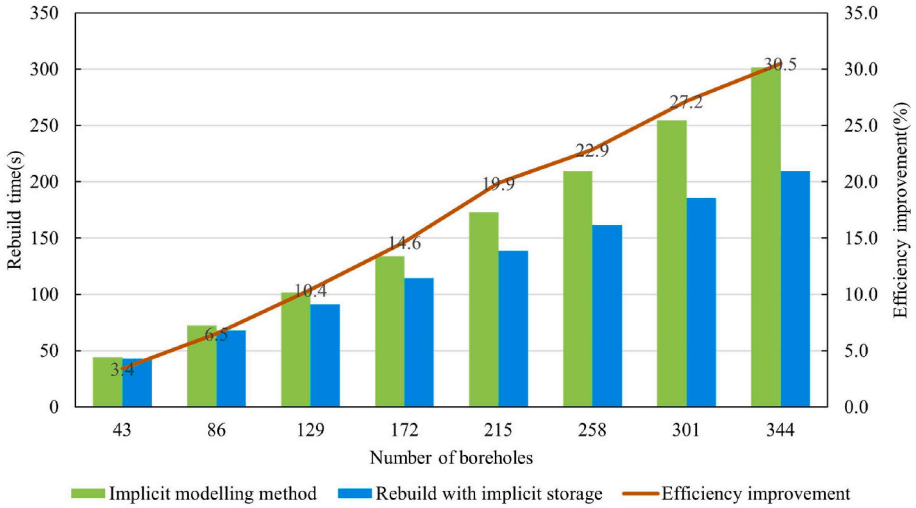


Fig. 13. Comparison of modelling time and reconstruction time from implicit storage with varying numbers of boreholes at a fixed modelling resolution.

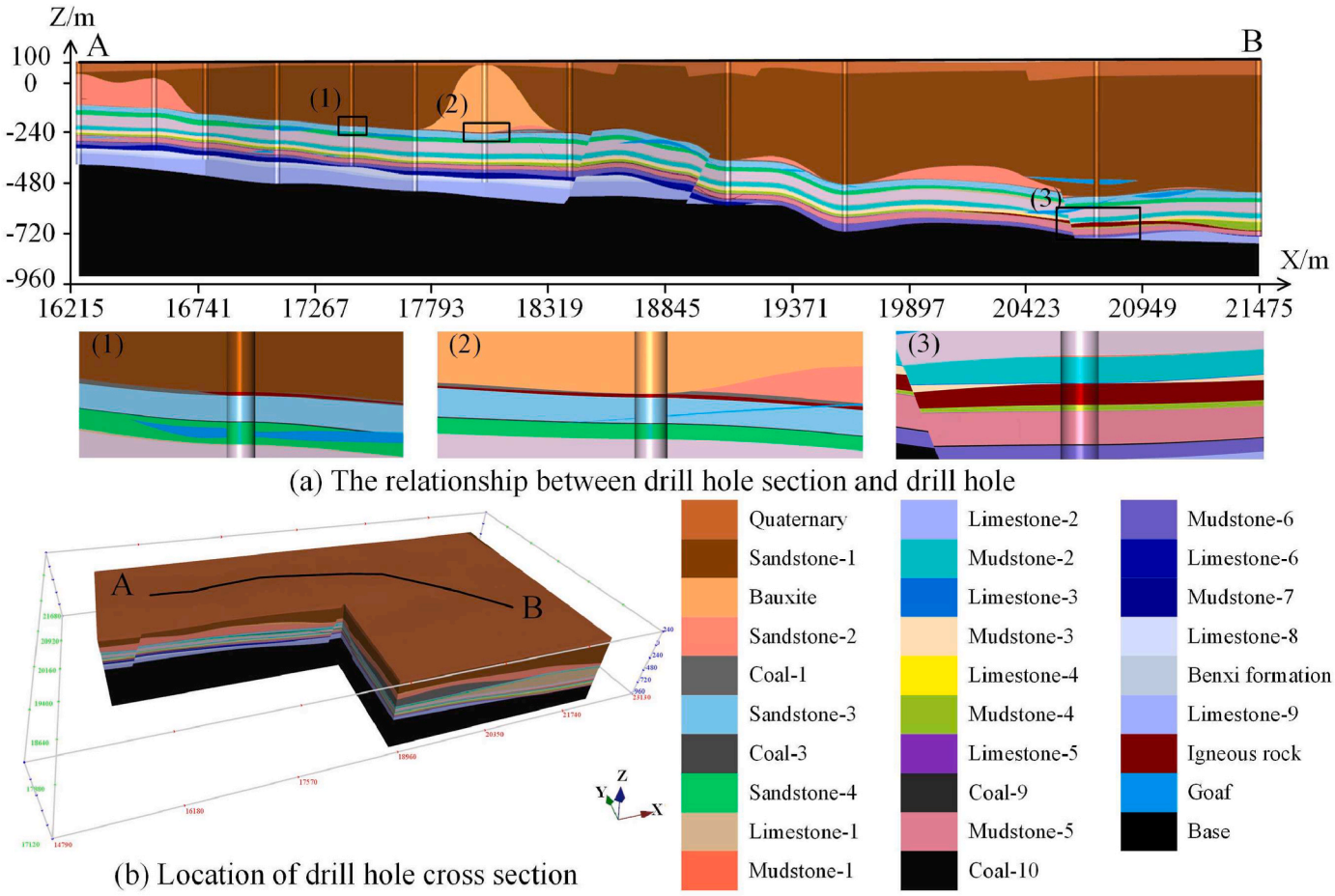


Fig. 14. Model results of the borehole line cross-section.

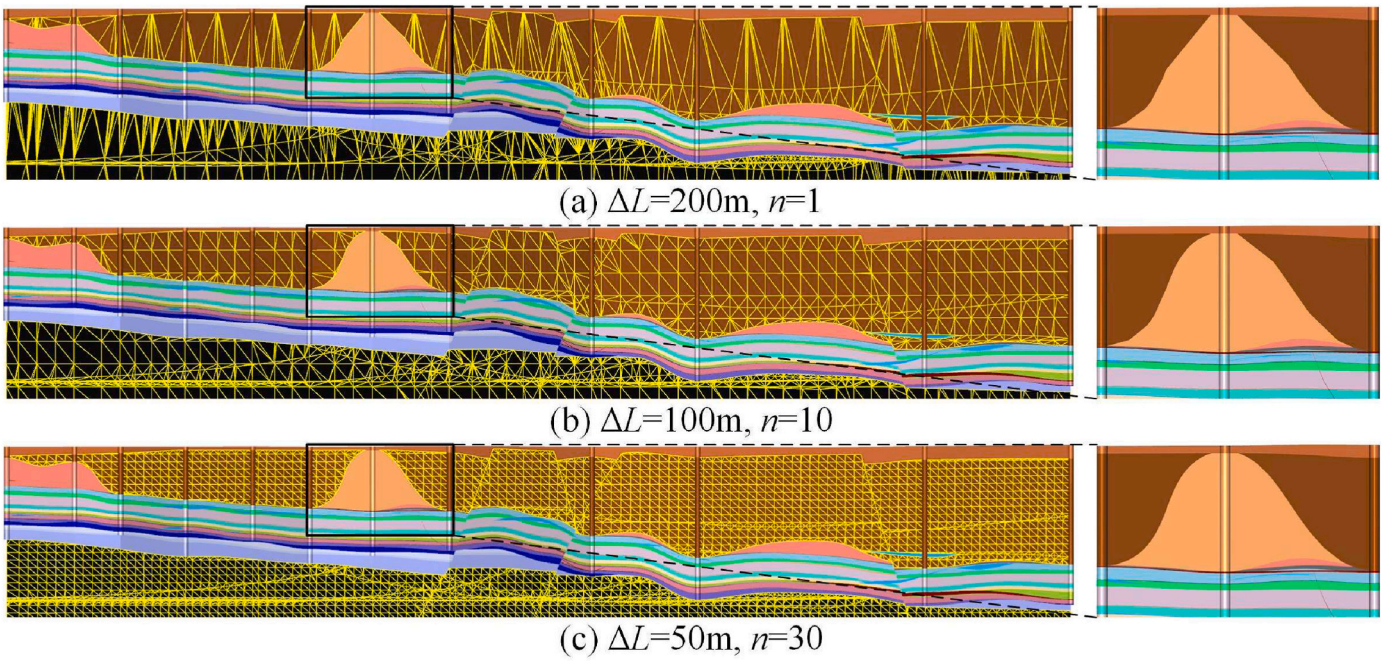


Fig. 15. The effects of different ΔL and n values on the section.

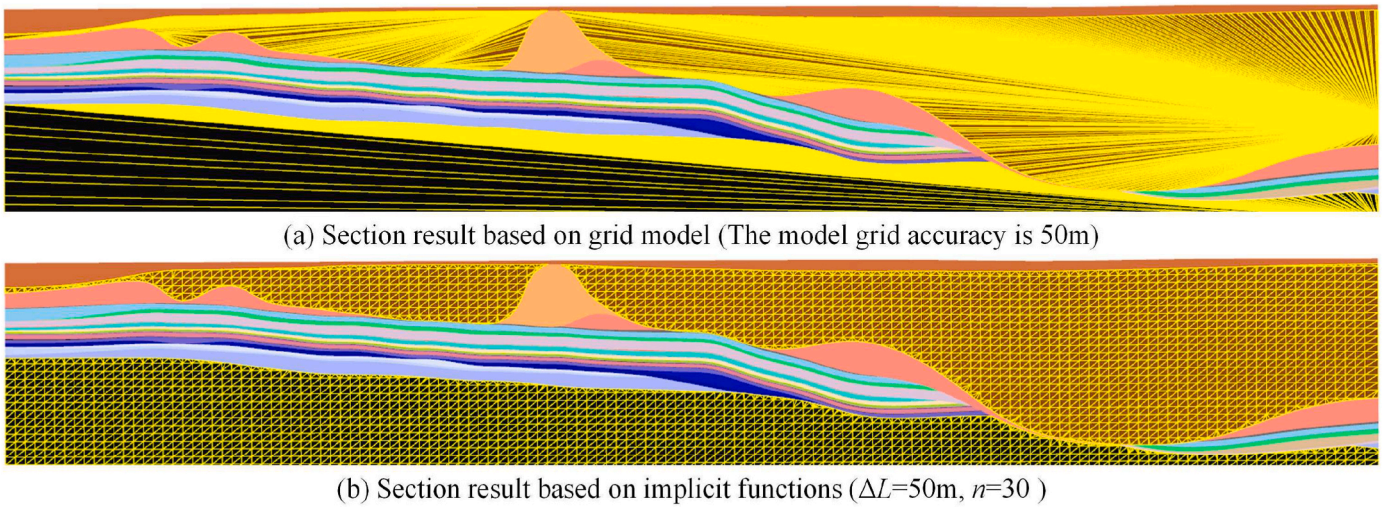


Fig. 16. Explicit and implicit section grids.

Table 2
Efficiency comparison between the implicit cross-sectioning method and the mesh-based explicit method.

Mesh scale (m)	Method	Cut 1 (s)	Cut 2 (s)	Cut 3 (s)	Cut 4 (s)	Cut 5 (s)	Average (s)	Improvement (%)
100	Explicit	22.119	22.014	22.111	22.019	22.180	22.089	98.7
	Implicit	0.300	0.294	0.302	0.292	0.293	0.296	
50	Explicit	47.378	47.774	47.544	47.675	47.891	47.652	98.8
	Implicit	0.576	0.560	0.568	0.567	0.562	0.567	

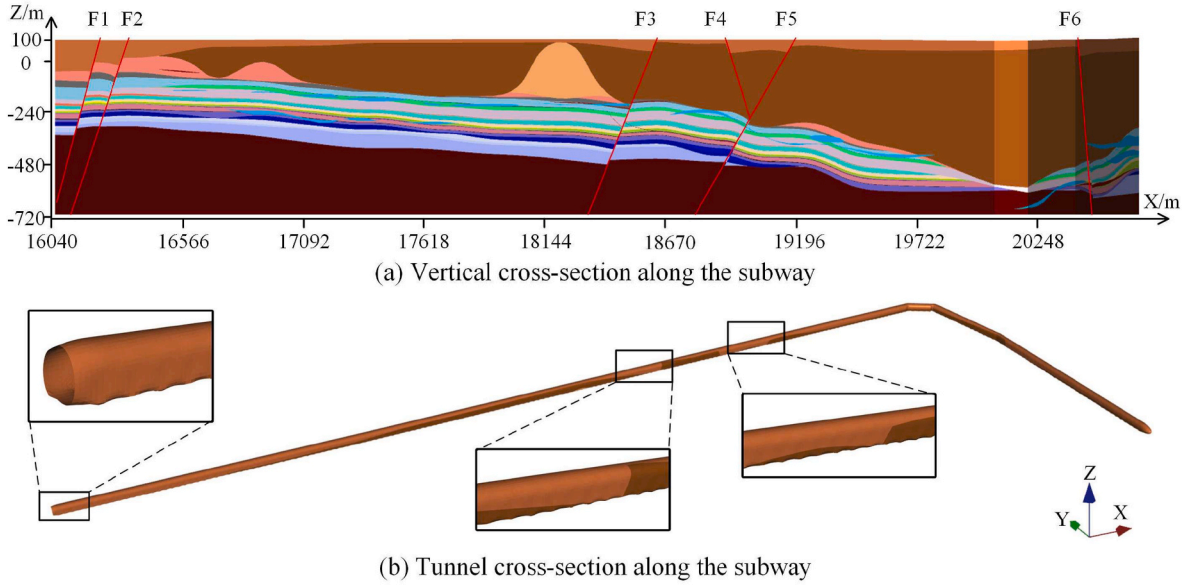


Fig. 17. Cross-section of the tunnel along the subway.

Traditional geological modelling approaches rely on explicit representations of models, which are stored as mesh-based data structures. When multiscale modelling is needed, users must construct and store a separate mesh model for each target resolution, which significantly increases both modelling time and storage overhead, and undermines the potential of implicit modelling for multiresolution applications. Our method allows for on-demand reconstruction of model geometry at any mesh resolution, which greatly reduces storage requirements and improves reconstruction efficiency, as demonstrated by our experiments across varying mesh scales and data volumes. In terms of cross-sectional analysis, traditional sectioning methods operate directly on explicit meshes, which are prone to numerical instability in the presence of topological errors. The implicit sectioning method leverages the analytical continuity of implicit surfaces to robustly intersect with cross-sectional geometries. This not only enhances computational stability and cross-section generation efficiency but also enables the generation of complex section geometries such as curved, tunnel-shaped, or function-defined profiles.

The proposed method should also be distinguished from classical meshing approaches, geometry-processing libraries such as CGAL, and

surface extraction algorithms such as Marching Cubes and Marching Tetrahedra. These methods mainly operate after the target geometric representation or scalar field has been defined. Classical meshing and geometry-processing methods are designed to generate, optimise, repair, clip, or process explicit geometric objects such as point sets, polygonal surfaces, tetrahedral meshes, or volumetric domains. Similarly, Marching Cubes and Marching Tetrahedra are commonly used to convert sampled implicit scalar fields into explicit meshes at the final stage of implicit modelling. These methods are effective and can still be regarded as downstream tools in the proposed workflow for mesh extraction, optimisation, or further geometric processing. By contrast, the focus of this study is on preserving the implicit modelling result before explicit mesh extraction, so that model reconstruction and cross-section generation can be performed on demand at different resolutions. In this sense, the proposed method does not replace existing meshing or geometry-processing techniques, but provides a storage and analysis framework that allows them to be integrated more flexibly into multiscale implicit geological modelling.

The proposed implicit storage and sectioning methods show strong potential for future applications. For example, in large-scale and data-intensive implicit 3D geological modelling, these methods can significantly improve the efficiency of multiscale modelling and analysis. The case study in this paper demonstrates that the efficiency gains in model reconstruction become more pronounced as the number of boreholes increases. Therefore, we conducted a preliminary test on a modelling case in Shenyang (Wang et al., 2024). The dataset included 28,000 boreholes and 75 stratigraphic layers, and covered an area of 455 km^2. Under a mesh resolution of 250 m, the full implicit modelling workflow took 1144.75 min, whereas with implicit storage, model reconstruction at the same resolution required only 145.62 min. This finding that the implicit surface computation accounted for as much as 87% of the total modelling time, highlighting the substantial time-saving potential of implicit storage in large-scale, multiresolution modelling. For large-area implicit 3D geological model visualization, future work could integrate LOD (level of detail) techniques with block-based modelling strategies to enable dynamic, real-time multiscale rendering. Even without the application of advanced LOD techniques, this method still enables efficient generation of multiple-resolution models, significantly reducing the processing time.

Despite the strengths demonstrated by the proposed method in various aspects, several limitations remain. First, in terms of the diversity of implicit functions, this study implemented only the HRBF, while other surface interpolation approaches—such as signed distance functions and deep learning-based models—could be incorporated to expand applicability. Second, regarding the storage format, the structure proposed in this study was designed solely to support the implemented method and has not yet undergone thorough validation or standardisation. Future work should further investigate how the method-specific metadata proposed in this study can be mapped to, or linked with, existing geoscience data exchange standards to improve interoperability. Further efforts are needed from the research community to improve and formalise storage standards for implicit surfaces. Moreover, the software implemented in this study requires the implicit function expressions to be embedded within the code, which limits flexibility; future research is needed to explore externalised expression handling to improve the method's generalisability. Another issue worth noting is that implicit cross-sectioning is tightly coupled with implicit storage, meaning that cross-sections can be generated only from stored implicit functions; as such, traditional mesh-based models cannot benefit from this method.

6. Conclusions

By storing the implicit functions of various interfaces in the implicit modelling process, this study developed a fast model reconstruction method tailored for multiscale modelling requirements. Efficiency comparisons are conducted from two perspectives. The experimental results show that the proposed method significantly reduces the time required to compute implicit functions when it is applied to multiscale modelling of the same dataset. Moreover, the improvement in efficiency is more significant with larger volumes of modelling data, further validating the method's potential for large-scale, multiresolution geological modelling. In addition, a cross-section generation method was proposed on the basis of the implicit storage approach, which enables direct generation of cross-sections from raw input data. The experimental results show that the proposed implicit cutting method not only achieves significant improvements in efficiency but also enhances applicability and support for diverse section geometries.

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CRediT authorship contribution statement

Zhibin Liu: Conceptualization, Data curation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Jiateng Guo:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Project administration, Validation, Writing – review & editing. **Xulei Wang:** Conceptualization, Methodology, Software, Validation, Writing – review & editing. **Luyuan Wang:** Methodology, Visualization, Writing – review & editing. **Jianqiao Zhang:** Data curation, Project administration, Resources, Supervision. **Xingzhou Zhang:** Data curation, Supervision, Validation. **Lixin Wu:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cageo.2026.106204.

Data availability

The implicit storage and cross-sectioning tools developed in this study were implemented using C++ and the Qt application framework. The developed model storage and retrieval software, along with its source code and test cases, is publicly available at the following repository: https://github.com/0620-bot/implicit-storage-3d-geological-models.

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